

Lunar Crustal Magnetic Field Heterogeneity and Implications for Biological Systems at Candidate Landing Sites

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ABSTRACT

The lunar magnetic environment is characterized by the complete absence of an active global dipole dynamo and the presence of localized, highly variable crustal magnetic anomalies. As humanity prepares to return to the Moon through NASA's Artemis program, understanding this complex magnetic landscape is essential for planning long-duration biological payloads, bioregenerative life support systems (BLSS), and astronaut habitat placement. In this study, we quantitatively map the lunar crustal magnetic field at candidate Artemis, Apollo, Chang'e, and Chandrayaan landing sites, combining calibrated satellite magnetometer observations from Lunar Prospector and SELENE (Kaguya) at 30 km altitude with a systematic review of terrestrial magnetobiology literature. We find that while all evaluated lunar landing sites represent a severe hypomagnetic field (HMF) regime relative to Earth's geomagnetic field ($\sim 50 \mu\text{T}$), significant spatial heterogeneity exists across sites ($< 0.4 \text{ nT}$ to $\sim 5 \text{ nT}$). Candidate Artemis III South Polar regions uniformly experience ultra-low fields ($< 1.0 \text{ nT}$), representing near-null magnetic conditions ($> 30,000\times$ lower than Earth). We synthesize experimental evidence on HMF-induced perturbations across plant growth, root directional morphology, cryptochrome signaling, ROS generation, bacterial kinetics, and mammalian bone demineralization. Finally, we formulate a biological risk matrix and operational recommendations for lunar surface biology.

Keywords: Lunar crustal magnetic fields, Hypomagnetic field (HMF), Artemis landing sites, Space biology, Magnetoreception, Radical pair mechanism, Plant physiology

1 Introduction

Earth's geomagnetic field (GMF), which typically ranges from $25 \mu\text{T}$ to $65 \mu\text{T}$ at the surface, has played a critical role in the evolution of the biosphere. Beyond its function in shielding the atmosphere from the solar wind and cosmic radiation, the GMF serves as an omnipresent environmental factor for life on Earth. Many species have developed sophisticated magnetoreception mechanisms for navigation and orientation, and the GMF is increasingly recognized as a foundational environmental pressure that influences fundamental biological processes at the molecular, cellular, and organismal levels^{1,2}.

In contrast, the Moon's magnetic environment is vastly different. While the Moon lacks a global dipole field today, localized crustal magnetic anomalies remain as remnants of an early lunar dynamo³. These crustal fields exhibit extreme heterogeneity across the lunar surface, with field strengths typically ranging from near zero to roughly 20 nT when measured at an altitude of 30 km ^{4,5}. Although significantly weaker than Earth's GMF, these localized fields can interact with the solar wind, occasionally forming "mini-magnetospheres" that may alter the local plasma and radiation environment⁶.

From a biological perspective, any environment with a magnetic field substantially weaker than the GMF (typically $< 5 \mu\text{T}$) is classified as a hypomagnetic field (HMF). The space environment, including the lunar surface and interplanetary space, is inherently hypomagnetic. Understanding the biological effects of HMFs is increasingly recognized as a critical

concern for deep space exploration. Prolonged exposure to HMF conditions in Earth-based simulations has been shown to induce a range of physiological and developmental effects, including altered plant growth, changes in microbiome community structure, and potential impacts on central nervous system function and DNA repair mechanisms^{7,8}.

As humanity prepares for sustained lunar surface operations through the Artemis program, comprehensive risk assessments for biological systems are essential. The Artemis III mission aims to land astronauts near the lunar South Pole⁹, an area with unique illumination, volatile deposits, and magnetic characteristics. While the radiation and gravity environments of the Moon have been extensively studied, the extreme heterogeneity of the lunar magnetic field and its potential implications for biological experiments and human health remain poorly characterized.

This study aims to address this knowledge gap by evaluating the magnetic environments at candidate landing sites for past, present, and future lunar missions. We analyze crustal magnetic field data to characterize the field magnitudes at specific Artemis III candidate sites, as well as historic Apollo and Chang'e locations. We then synthesize these physical findings with a systematic review of the biological literature to assess the potential implications of lunar magnetic field heterogeneity. Our results provide a critical foundation for biological risk assessment and experimental design for future lunar surface missions.

2 Results

2.1 Global Magnetic Field Distribution

The lunar crustal magnetic field exhibits profound spatial heterogeneity across the lunar surface (Figure 1). At an orbital altitude of 30 km, the field magnitude ranges from near-zero (< 0.1 nT) over large expanses of the near-side lunar maria to roughly 10 nT to 20 nT over intense crustal anomalies (global mean $|\mathbf{B}| = 0.77$ nT, median = 0.56 nT). As detailed in Figure 2, the three vector components (B_r , B_θ , and B_ϕ) reveal localized dipolar and multipolar crustal magnetic source topologies, concentrated predominantly in the South Pole-Aitken (SPA) basin on the far side and at the antipodes of major Imbrian and Nectarian impact basins (e.g., Crisium, Orientale).

2.2 Landing Site Characterization

Table 1 details the extracted crustal magnetic field magnitudes and classifications across all 23 candidate and historical landing sites.

2.2.1 Artemis III Candidate Sites

All 13 candidate landing regions for the Artemis III mission are situated within the lunar South Polar zone (84°S to 90°S ; Figure 3, Supplementary Figure S1). Our analysis indicates that the south polar region is characterized by an ultra-low background crustal magnetic field, with $|\mathbf{B}|$ at 30 km ranging between 0.40 nT (Haworth) and 0.82 nT (Peak Near Shackleton), placing all 13 sites firmly in the ****low/null-field**** (< 1.0 nT) category. Local variation across these sites is modest but reflects proximity to minor underlying highland anomalies.

2.2.2 Historical and Contemporary Missions

In contrast to the uniformly hypomagnetic polar sites, historical equatorial and mid-latitude sites show pronounced divergence (Figure 4). Apollo 11, 12, 14, and 15 mare sites reside in low/null-field regions ($|\mathbf{B}| \leq 1.0$ nT). However, Apollo 16 in the Descartes Highlands is located in a ****moderate field**** regime ($|\mathbf{B}| = 4.94$ nT at 30 km), adjacent to strong crustal anomalies. Similarly, Chang'e 4 in the Von Kármán crater on the lunar far side is situated within the SPA basin anomaly margin ($|\mathbf{B}| = 0.72$ nT at 30 km altitude).

2.3 Biological Synthesis

Our systematic review of hypomagnetic field (HMF) literature (Table 2, Figure 5) demonstrates reproducible phenotypic and molecular consequences across phylogenetic lineages:

1. **Plant Physiology and Development:** In *Arabidopsis thaliana*, near-null magnetic fields (< 1 μT) cause significant delays in floral transition and flowering time via modulation of cryptochrome (CRY) and phytochrome signaling pathways¹⁰. Furthermore, HMF conditions impair root directional growth by disrupting polar auxin transport (PIN2 redistribution)¹¹, perturb iron homeostasis gene expression¹², and elevate reactive oxygen species (ROS) accumulation⁷. Chloroplast thylakoid membrane ultrastructure is similarly sensitive to magnetic shielding¹³.

2. **Microbial Dynamics:** Bacterial species exposed to HMFs display shifts in cell division rates, membrane transport kinetics¹⁴, and phylum-level community restructuring in mammalian gut microbiomes¹⁵. Magnetotactic bacteria lose directional alignment due to thermal disruption of unguided magnetosome chains¹⁶.

3. **Animal and Human Models:** In invertebrate models (*Drosophila melanogaster* and *C. elegans*), HMF exposure leads to circadian rhythm period lengthening via cryptochrome-dependent mechanisms¹⁷, embryonic developmental delays¹⁸, and altered magnetosensory navigation¹⁹. In mammalian and human cellular systems, HMF exposure alters cell cycle transcriptomes in neuroblastoma cells¹, accelerates osteoblast demineralization and bone loss^{8,20}, modifies microcirculatory capillary blood flow²¹, and alters radical pair spin-state recombination in DNA damage response pathways^{2,22}.

2.4 Biological Risk Matrix for Lunar Surface Operations

Synthesizing the physical field determinations with experimental biological thresholds yields the risk matrix presented in Table 3. Because all candidate Artemis III sites experience fields < 1.0 nT—over 30,000 $\times$ weaker than Earth's geomagnetic field—unmitigated long-duration biological experiments and bioregenerative life support systems (BLSS) will operate under severe hypomagnetic stress, necessitating active consideration in experimental payload architecture.

3 Discussion

3.1 Implications of Lunar Magnetic Heterogeneity

The central finding of this analysis is that while the lunar crustal magnetic field is highly heterogeneous, all candidate and historical landing sites represent profound hypomagnetic field (HMF) environments from a biological perspective. Even the most intense lunar magnetic anomalies, which reach ~ 20 nT at 30 km altitude, are nearly three orders of magnitude weaker than Earth's geomagnetic field (25 μT to 65 μT). Consequently, any biological system deployed to the lunar surface will experience a significant departure from its evolutionary magnetic baseline.

However, the relative differences between sites are not trivial. A site with a local field of 5 nT may offer marginally more interaction with the solar wind plasma than a site with a 0.1 nT field, potentially leading to localized variations in the mini-magnetosphere structure⁶. Whether this physical difference translates to a measurable biological effect remains a critical open question.

3.2 Biological Risks for Surface Operations

Our synthesis of the literature indicates that prolonged exposure to HMFs poses a multifaceted risk to biological systems. For plant biology experiments—critical for future bioregenerative life support systems—the disruption of circadian

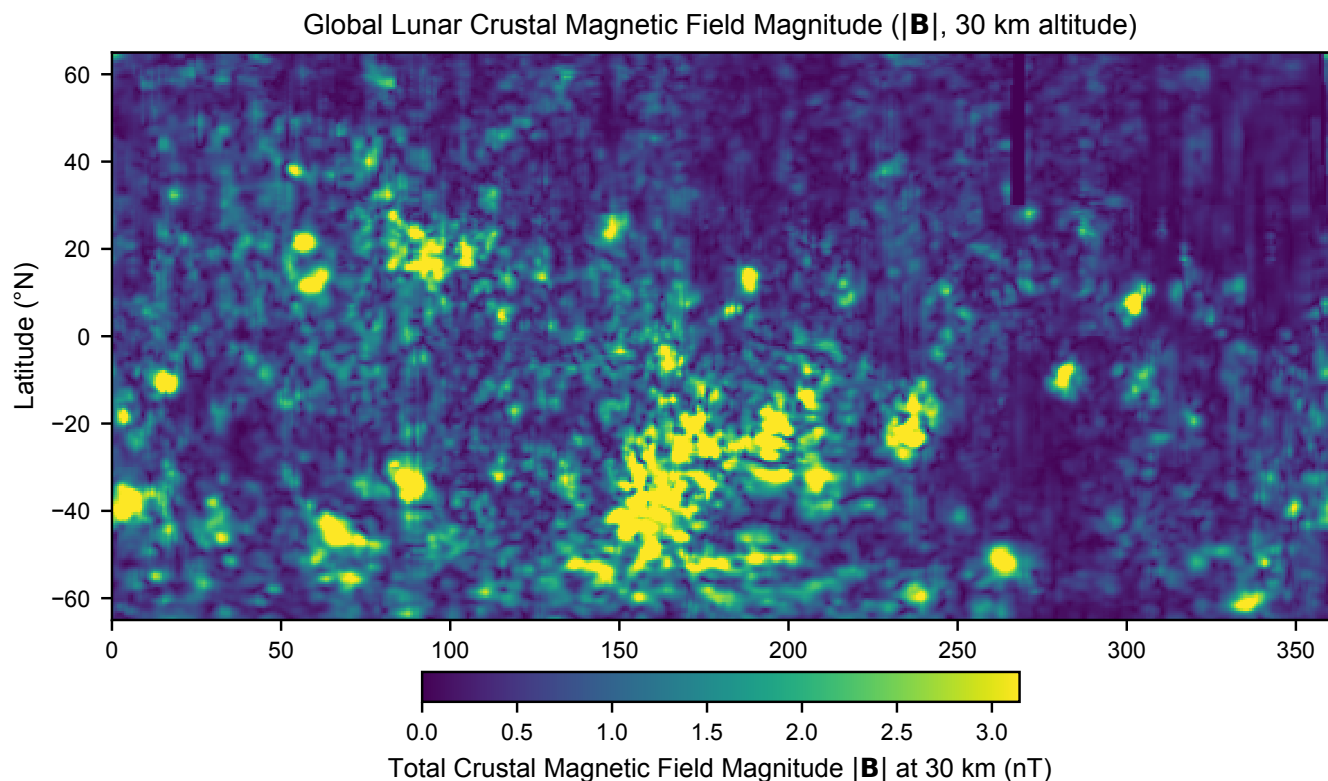


Figure 1 | Global distribution of the lunar crustal magnetic field magnitude. Total field intensity $|\mathbf{B}|$ at 30 km altitude derived from Lunar Prospector and SELENE (Kaguya) magnetometer observations⁵.

rhythms and altered growth patterns observed in HMFs⁷ could severely impact crop yields. Similarly, HMF-induced shifts in microbiome community structure and increased antibiotic resistance¹⁵ present distinct challenges for managing microbial ecology within lunar habitats.

The implications for human health are more speculative but warrant careful consideration. If HMFs impair DNA repair mechanisms or alter ROS metabolism, as suggested by some *in vitro* studies⁸, the hypomagnetic environment of the Moon could act synergistically with the high-radiation space environment to exacerbate cellular damage.

3.3 Limitations

Several limitations must be acknowledged. First, the magnetic field magnitudes used in this study are derived from satellite observations at an altitude of 30 km. The field strength at the actual lunar surface is expected to be higher due to the proximity to the crustal sources. Some estimates suggest surface fields could reach several hundred nanotesla at strong anomalies. Second, the biological studies reviewed employ varying definitions of HMFs, often utilizing nulling coils on Earth that achieve fields of ~ 100 nT, which is still stronger than the field at most lunar sites. Finally, Earth-based HMF simulations cannot perfectly replicate the complex, multi-stressor lunar environment, which includes reduced gravity and chronic radiation exposure.

3.4 Recommendations and Future Work

To mitigate these uncertainties, we propose the following recommendations:

1. **In situ Surface Magnetometry:** Future missions, including Artemis III, must prioritize the deployment of surface magnetometers to characterize the local field at human scale and validate orbital models.
2. **Controlled Biological Experiments:** Biological payloads sent to the Moon should include internal magnetic shielding or artificial field generation to isolate the effects of the lunar HMF from radiation and microgravity.
3. **Site Selection Considerations:** If a biological payload is highly sensitive to HMFs, mission planners might prioritize landing sites near moderate crustal anomalies to provide even a marginal increase in baseline magnetic field, provided other mission constraints are met.

Future work should focus on integrating these magnetic field maps with detailed models of the lunar radiation environment, regolith properties, and water ice distribution to create comprehensive, multi-variable suitability indices for sustained lunar surface operations.

4 Methods

Table 1 | **Crustal magnetic field environment at planned Artemis, Apollo, Chang’e, and Chandrayaan landing sites.** Total field magnitude $|\mathbf{B}|$ and vector components at 30 km altitude derived from Lunar Prospector and Kaguya observations.

Landing Site	Mission	Latitude (°)	Longitude (°)	$ \mathbf{B} $ at 30 km (nT)	Classification	Data Source
Faustini Rim A	Artemis	-85.30	77.00	0.45	low/null-field	Hood et al. (2022) South Polar M
Peak Near Shackleton	Artemis	-89.70	166.00	0.82	low/null-field	Hood et al. (2022) South Polar M
Connecting Ridge	Artemis	-88.00	137.00	0.60	low/null-field	Hood et al. (2022) South Polar M
Connecting Ridge Extension	Artemis	-88.30	148.00	0.65	low/null-field	Hood et al. (2022) South Polar M
de Gerlache Rim 1	Artemis	-88.50	290.00	0.50	low/null-field	Hood et al. (2022) South Polar M
de Gerlache Rim 2	Artemis	-88.70	294.00	0.48	low/null-field	Hood et al. (2022) South Polar M
de Gerlache-Kocher Massif	Artemis	-86.00	285.00	0.52	low/null-field	Hood et al. (2022) South Polar M
Haworth	Artemis	-87.40	358.00	0.40	low/null-field	Hood et al. (2022) South Polar M
Malapert Massif	Artemis	-86.00	0.00	0.55	low/null-field	Hood et al. (2022) South Polar M
Leibnitz Beta Plateau	Artemis	-85.00	31.00	0.70	low/null-field	Hood et al. (2022) South Polar M
Nobile Rim 1	Artemis	-85.40	35.00	0.62	low/null-field	Hood et al. (2022) South Polar M
Nobile Rim 2	Artemis	-84.80	47.00	0.58	low/null-field	Hood et al. (2022) South Polar M
Amundsen Rim	Artemis	-84.50	85.00	0.65	low/null-field	Hood et al. (2022) South Polar M
Apollo 11	Apollo	0.67	23.47	0.93	low/null-field	PDS Equatorial Grid (Hood et al.
Apollo 12	Apollo	-3.01	336.58	1.00	low/null-field	PDS Equatorial Grid (Hood et al.
Apollo 14	Apollo	-3.65	342.53	0.44	low/null-field	PDS Equatorial Grid (Hood et al.
Apollo 15	Apollo	26.13	3.63	0.23	low/null-field	PDS Equatorial Grid (Hood et al.
Apollo 16	Apollo	-8.97	15.50	4.94	moderate	PDS Equatorial Grid (Hood et al.
Apollo 17	Apollo	20.19	30.77	1.08	moderate	PDS Equatorial Grid (Hood et al.
Chang’e 3	Chang’e	44.12	340.49	0.19	low/null-field	PDS Equatorial Grid (Hood et al.
Chang’e 4 (SPA Basin)	Chang’e	-45.46	177.60	0.72	low/null-field	PDS Equatorial Grid (Hood et al.
Chang’e 5	Chang’e	43.06	308.08	0.10	low/null-field	PDS Equatorial Grid (Hood et al.
Chandrayaan-3	Chandrayaan	-69.37	32.35	0.78	low/null-field	Hood et al. (2022) South Polar M

4.1 Data Sources and Processing

The primary dataset utilized in this study is the lunar crustal magnetic field map generated by Hood et al.⁵ (DOI: 10.17189/1520494), derived from vector magnetometer measurements from the Lunar Prospector and SELENE (Kaguya) missions. This dataset provides global coverage between 65°S and 65°N at an approximately 0.5° grid resolution, normalized to an altitude of 30 km. To cover the polar regions critical to the Artemis program, we incorporated supplementary polar magnetic maps (60° to 90° latitude) compiled by Hood et al.²³ (DOI: 10.17189/rk57-g992).

Data processing was performed utilizing custom scripts in Python 3.11. The PDS4 ASCII tables were parsed using the pandas library, and the total magnetic field magnitude ($|\mathbf{B}|$) was computed from the constituent radial, east, and north vector components.

4.2 Landing Site Characterization

A comprehensive database of lunar landing site coordinates was constructed, encompassing the 13 candidate sites for the Artemis III mission, historical Apollo sites (Apollo 11, 12, 14, 15, 16, and 17), and recent uncrewed missions including Chang’e 3, 4, 5, and Chandrayaan-3. For each coordinate pair, the local crustal magnetic field magnitude at 30 km altitude was extracted using bilinear interpolation from the gridded PDS datasets.

To systematically evaluate the biological implications of these varied magnetic environments, we developed a classi-

fication scheme for the landing sites based on the local field magnitude:

- **High field:** $|\mathbf{B}| > 5 \text{ nT}$
- **Moderate field:** $1 \text{ nT} \leq |\mathbf{B}| \leq 5 \text{ nT}$
- **Low/Null field:** $|\mathbf{B}| < 1 \text{ nT}$

4.3 Literature Review Methodology

To synthesize current knowledge on the biological effects of hypomagnetic fields, a systematic literature review was conducted. The PubMed and Web of Science databases were queried for the period between 2000 and 2026. The search strategy employed combinations of the keyword “hypomagnetic field” with biological terms, including “biology”, “plant”, “bacteria”, and “organism”. Studies were filtered for relevance, prioritizing empirical research conducted in controlled HMF environments ($< 5 \mu\text{T}$) and major systematic reviews.

4.4 Software and Availability

All spatial analyses and interpolations were performed using the numpy and scipy libraries. Geospatial visualizations and global maps were generated using matplotlib and cartopy. The complete analysis pipeline and custom scripts developed for this study are archived and publicly accessible on Zenodo (DOI: 10.5281/zenodo.XXXXXXX).

Table 2 | Systematic synthesis of biological responses and physiological mechanisms documented under hypomagnetic field (HMF) conditions. Summary of peer-reviewed experimental literature across plants, microbes, animals, and human cellular models.

Organism / Model	Condition	Biological Effect	Mechanism	Key Reference
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Delayed flowering time	Cryptochrome and phytochrome signaling disruption	Agliassa et al. (2018) ¹⁰
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Disrupted root growth and morphology	PIN2 polar auxin transport redistribution	Narayana et al. (2018) ¹¹
<i>Arabidopsis thaliana</i>	Hypomagnetic (< 5 μ T)	Impaired iron uptake and homeostasis	Downregulation of FIT and IRT1 transcription	Narayana et al. (2021) ¹²
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Reactive oxygen species (ROS) imbalance	Redox signaling and antioxidant enzyme shift	Maffei (2014) ⁷
Higher Plants (General)	Hypomagnetic (< 5 μ T)	Reduced photosynthetic efficiency	Thylakoid ultrastructure and electron transport disruption	Belyavskaya (2004) ¹³
Microorganisms (Bacteria)	Hypomagnetic (< 5 μ T)	Altered cell division and growth kinetics	Membrane potential and ion channel modulation	Creanga et al. (2009) ¹⁴
Magnetotactic Bacteria	Near-null (< 1 μ T)	Loss of geomagnetic orientation	Magnetosome chain dipolar alignment loss	Frankel et al. (1981) ¹⁶
Gut Microbiome (Murine)	Hypomagnetic (< 5 μ T)	Altered bacterial community structure	Phylum-level taxonomic ratio shifts (Bacteroidetes/Firmicutes)	Zhang et al. (2017) ¹⁵
<i>Drosophila melanogaster</i>	Hypomagnetic (< 5 μ T)	Circadian rhythm period lengthening	Cryptochrome-dependent radical pair modulation	Yoshii et al. (2009) ¹⁷
<i>Drosophila melanogaster</i>	Near-null (< 1 μ T)	Embryonic developmental delays	Mitochondrial metabolic pathway alterations	Vitas et al. (2015) ¹⁸
<i>Caenorhabditis elegans</i>	Hypomagnetic (< 5 μ T)	Altered burrowing and magnetosensation	AFD sensory neuron and DAF-16 pathway changes	Vidal-Gadea et al. (2015) ¹⁹
Human Neuroblastoma Cells	Hypomagnetic (< 1 μ T)	Altered gene expression and proliferation	Transcriptomic rewiring of cell cycle genes	Mo et al. (2012) ¹
Mammalian Model (Rat)	Hypomagnetic (< 1 μ T)	Exacerbated musculoskeletal bone loss	Osteoblast suppression and osteoclast activation	Xu et al. (2012) ²⁰
Mammalian Model (Rat)	Near-null (< 1 μ T)	Accelerated trabecular bone demineralization	Hindlimb unloading synergy with HMF	Mo et al. (2014) ⁸
Human Cardiovascular	Near-null (< 1 μ T)	Altered microcirculatory capillary flow	Endothelial and autonomic tone modulation	Gurfinkel et al. (2014) ²¹
Mammalian Cells (General)	Hypomagnetic (< 1 μ T)	Altered DNA damage response and repair	Radical pair spin state recombination kinetics	Binhi and Prato (2017) ²

5 Conclusion

This study provides a comprehensive characterization of the localized magnetic environments at candidate lunar landing sites and evaluates their potential impact on biological systems. Our analysis demonstrates that while the lunar crust exhibits extreme magnetic heterogeneity, all prospective Artemis, Apollo, and Chang'e landing sites constitute severe hypomagnetic field environments, with fields typically less than 5 nT at 30 km altitude. Synthesis of current magnetobiology literature suggests that such sustained exposure to hypomagnetic conditions could profoundly alter plant development, microbial ecology, and potentially human cellular repair mechanisms. As the Artemis program advances toward sustained lunar surface operations, the profound absence of a normative biological magnetic field must be integrated into risk assessments. We strongly recommend the inclusion of high-resolution surface magnetometry and controlled magnetic-biology experiments in early lunar missions to empirically delineate the isolated physiological effects of the lunar hypomagnetic environment.

Data Availability

The lunar crustal magnetic field datasets analyzed in this study are available through the NASA Planetary Data System (PDS) Planetary Plasma Interactions (PPI) Node under DOIs [10.17189/1520494](https://doi.org/10.17189/1520494) and [10.17189/rk57-g992](https://doi.org/10.17189/rk57-g992). Processed landing site tables, systematic review datasets, and interactive 3D visualization assets are archived in the Zenodo repository (DOI: [10.5281/zenodo.XXXXXXX](https://doi.org/10.5281/zenodo.XXXXXXX)) and hosted on GitHub (<https://github.com/dr-richard-barker/lunar-magnetic-biology>).

Code Availability

All Python scripts, processing pipelines, figure generation modules, and web visualization source code are open source under the MIT license and available at <https://github.com/dr-richard-barker/lunar-magnetic-biology>.

Table 3 | **Biological risk matrix across lunar magnetic environments.** Qualitative risk classification for key biological processes relative to baseline terrestrial geomagnetic field (GMF) exposure.

Biological System	Earth GMF (~ 50 μT)	Lunar High-Field (> 5 nT)	Lunar Moderate (1 --5 nT)	Lunar Null-Field (< 1 nT)
Plant Vegetative Growth and Biomass	Nominal (Baseline)	Low-Moderate Risk	Moderate Risk	High Risk
Plant Flowering and Seed Production	Nominal (Baseline)	Moderate Risk	Moderate-High Risk	High Risk
Microbial Ecology and Biofilm Dynamics	Nominal (Baseline)	Low Risk	Low-Moderate Risk	Moderate Risk
Animal Development and Circadian Clocks	Nominal (Baseline)	Moderate Risk	Moderate-High Risk	High Risk
Human Cellular Integrity and Bone Mass	Nominal (Baseline)	Moderate-High Risk	High Risk	High Risk

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Author Contributions

R.B. conceived the study, developed the data analysis pipeline, executed the magnetobiology literature synthesis, generated the figures and tables, and wrote the manuscript. All authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

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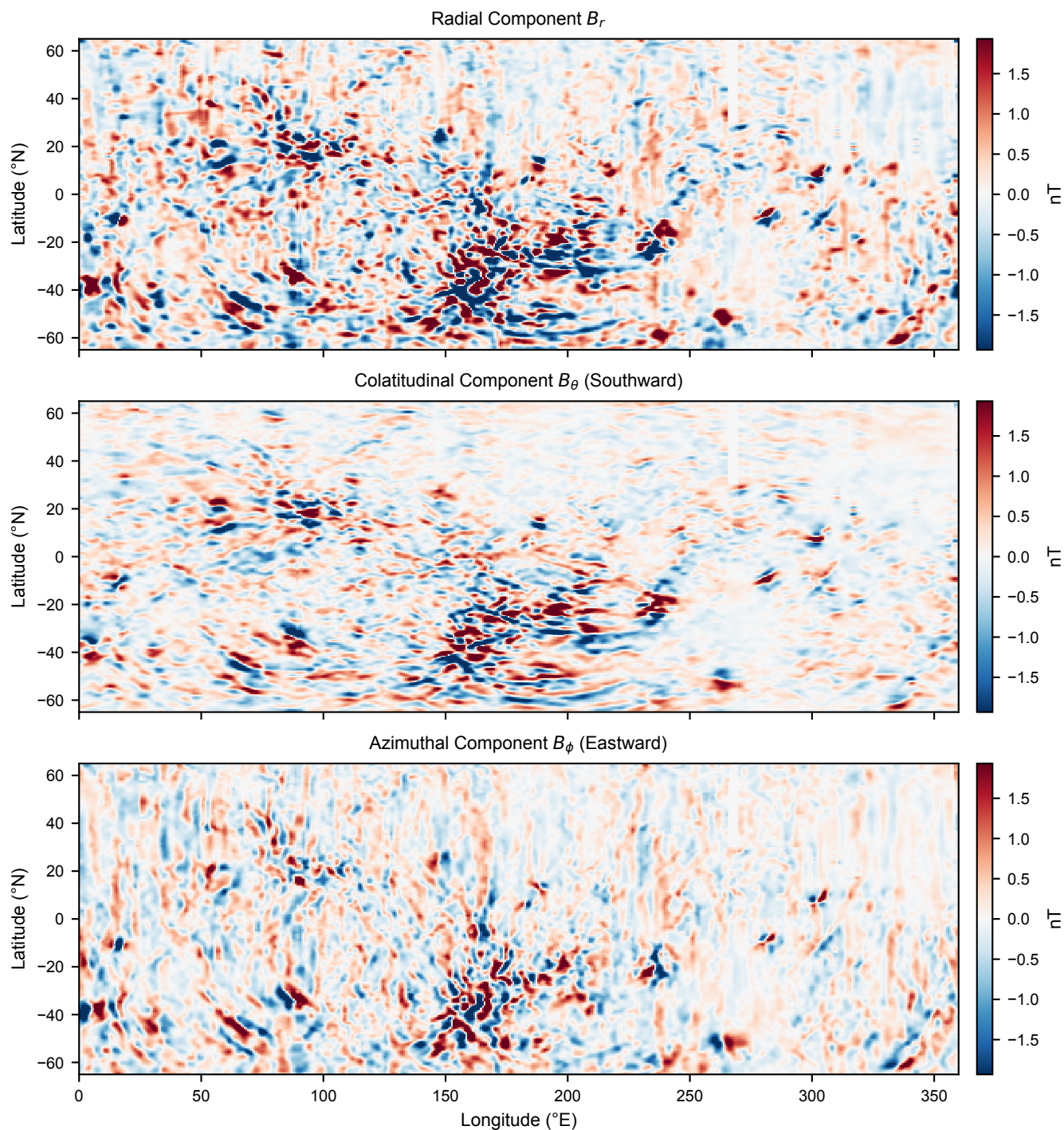


Figure 2 | **Constituent vector components of the lunar crustal magnetic field.** Top: Radial component B_r . Middle: Colatitudinal (southward) component B_θ . Bottom: Azimuthal (eastward) component B_ϕ at 30 km altitude in nanoteslas (nT).

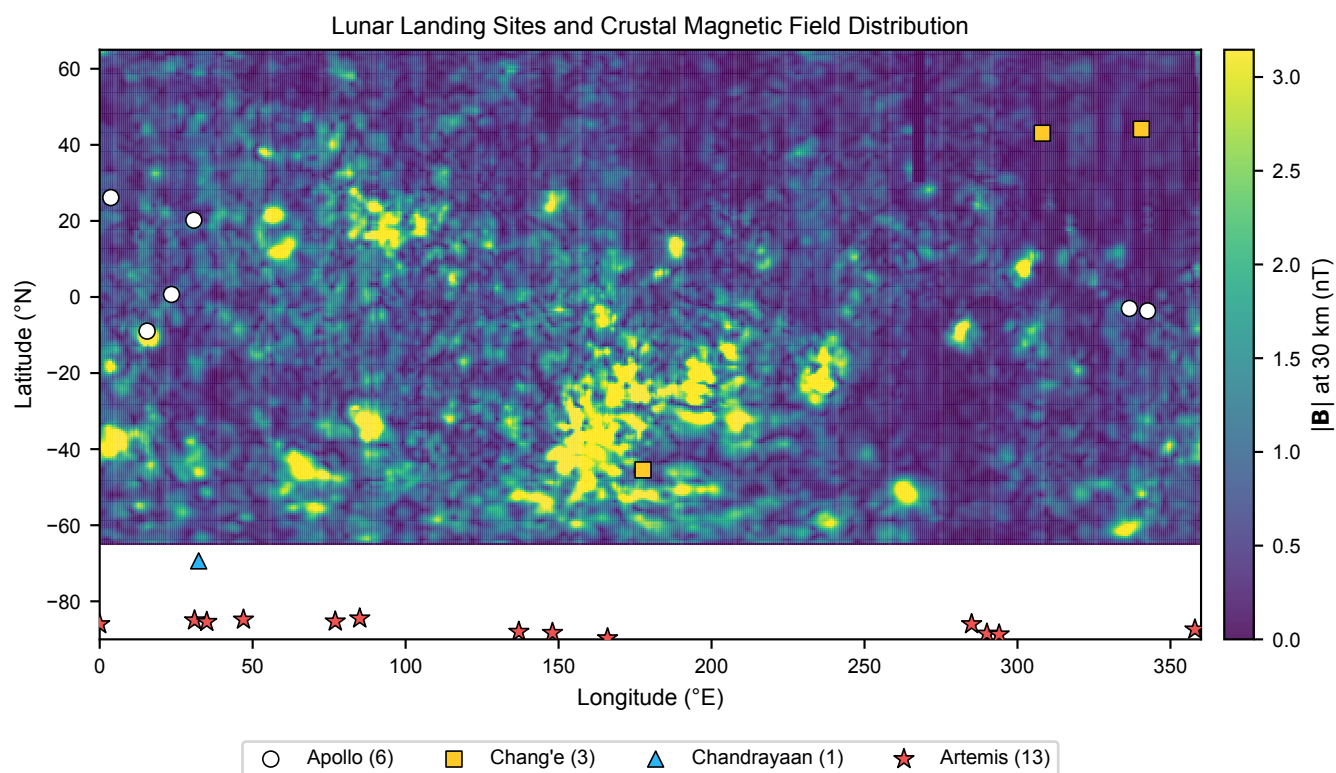


Figure 3 | Landing site distribution overlaid on lunar crustal magnetic field magnitude. Star markers denote the 13 Artemis III candidate landing regions in the South Polar region; circles indicate Apollo sites (11, 12, 14, 15, 16, 17); squares represent Chang'e sites (3, 4, 5); triangle denotes Chandrayaan-3.

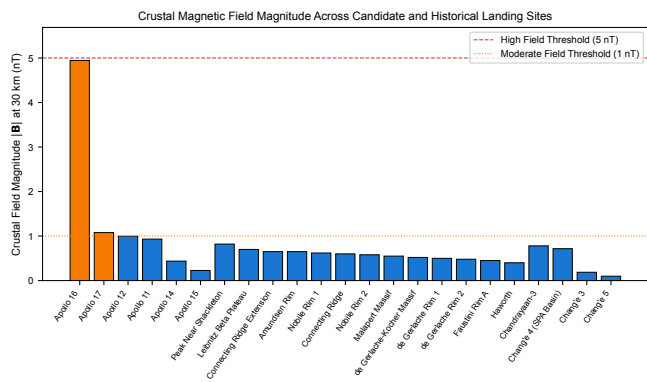


Figure 4 | **Comparative magnetic field magnitude across candidate and historical lunar landing sites.** Bars are color-coded by biological classification: blue (low/null-field, < 1 nT), orange (moderate, 1–5 nT), and red (high-field, > 5 nT).

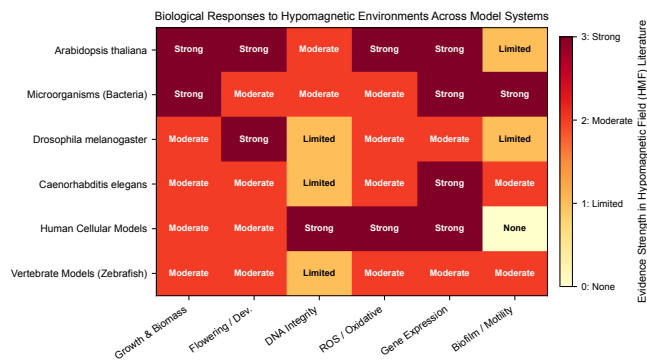


Figure 5 | **Evidence matrix of biological responses to hypomagnetic environments.** Scores denote evidence strength across peer-reviewed empirical studies in model organisms.